MTP Department
SIMHADRI

Note No :1

Ref:SMTPP/MAINT. PLNG./2025-26/MTP/420789

Date: 15/09/2025(16:31:27)

Subject: Approval of revision in LMI on Maintenance Optimization Through Predictive Maintenance

Sub: Approval of revision in LMI on Maintenance Optimization Through Predictive Maintenance

LMI/OIN/OPS/SYST/010, Rev. 04 and SEP 2025 is prepared based on revised COS-ISO-00-OIN/OPS/SYST/010, Rev. No.: 4 APR 2025

The following changes were incorporated in LMI as per revised OIN:

Clause 5.6.1: a) Added: Addition of Acoustic Imager details.

clause 5.5 has been corrected as discussed with EEMG:

Parameter	Frequency
Cooling tower eff. testing	once in a Year
Dirty air flow tests for coal flow	once in 6 months
APH Efficiency Test	once in 3 months

Submitted for review and furthur approval please

Submitted by:
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On:15/09/2025(16:31:27) **Ref:**SMTPP/MAINT. PLNG./2025-26/MTP/420789

Note No:2

Submitted for approval please.

Submitted by:

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Note No:3

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Unauthorized Use Prohibited

Note No:4

Reviewed

Submitted by:

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Note No:5

LMI has been modified, in line with the OIN. Forwarded for further review.

Submitted by:

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Note No:6

Reviewed and is being forwarded for further process please.

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Note No:7

The LMI is reviewed and found in line with practices.

Submitted for further processing of approval please.

Submitted by:

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Note No:8

Checked and forwarded.

Submitted by:

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Note No:9

Reviewed, forwarded for approval please.

Submitted by:

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Note No:10

LMI has been revised in line with the OIN. May please be approved.

Submitted by:

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On:29/09/2025(11:57:00) **Ref:**SMTPP/MAINT. PLNG./2025-26/MTP/420789

Note No:11

May please approve.

Submitted by:

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Note No:12

Submitted by:

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Note No:13

Approved & Submitted by:

समीर शर्मा - कार्यकारी निदेशक

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On:29/09/2025(17:40:33) **Ref:**SMTPP/MAINT. PLNG./2025-26/MTP/420789

NTPC **SIMHADRI**

LOCATION MANAGEMENT INSTRUCTION

Maintenance Optimization Through Predictive Maintenance

LMI/OIN/OPS/SYST/010 Rev: 04

SEP 2025

Approved for Implementation by HOP (SIMHADRI)

LMI APPROVAL DOCUMENTS

1	LMI NAME	Maintenance Optimization Through Predictive Maintenance
2	LMI NO.	LMI/OIN/OPS/SYST/010, Rev:04
3	Reference Document	COS-ISO-00-OIN/OPS/SYST/010, Rev. No: 4 APR 2025
4	Superseded Document	LMI-COS-ISO-00-OIN/OPS/SYST/010, Rev-3 Jun 2020
5	PREPARED BY:	SUSHILA, ENGINEER (MTP)
6	CHECKED BY:	KVK BHASKAR, AGM (MTP) DIVYA CHAWLA, AGM (MTP) MALAY KANT JANA, AGM (EMD) SANTANU JANA, DGM (CHEMISTRY) SOMNATH BHATTACHARYA, AGM (OPN) A.R. KOTESWARA RAO, AGM (FM)
7	REVIEWED BY:	AHMAD KAUNAIN RAZA, GM (O&M)
8	APPROVED BY:	SAMEER SHARMA, HOP-SIMHADRI

INDEX

TITLE	**Maintenance Optimization Through Predictive Maintenance**
LMI NO.	**LMI/OIN/OPS/SYST/010 Rev:04**

1.0 INTRODUCTION

The objectives of the Predictive Maintenance are

1.1 To increase the reliability and availability of equipment by monitoring and analyzing the operating conditions of equipment and their trends and thereby to reduce forced outage.

1.2 To reduce maintenance cost on account of breakdown/ forced outage, preventive maintenance and to reduce over/ under maintenance.

2.0 SUPERSEDED DOCUMENTS: LMI-COS-ISO-00-OIN/OPS/SYST/010, Rev-3

3.0 DETAILS OF REVISION:

Sl No.	Rev:03	Rev:04
1		Clause 5.6.1: a) Added: Addition of Acoustic Imager details

4.0 OBJECTIVE:

This LMI covers procedures for the following condition monitoring, analysis techniques and Performance monitoring through the testing and predictive maintenance program to be followed in the Station to optimize the maintenance strategies.

1) Vibration Monitoring and Analysis
2) Infrared Thermography
3) Dissolved Gas analysis of Transformers
4) Monitoring of Lube Oil Wear Debris Analysis
5) Major equipment Performance Monitoring
6) Acoustic Steam Leak Detection system in Boilers

TITLE	Maintenance Optimization Through Predictive Maintenance
LMI NO.	LMI/OIN/OPS/SYST/010 Rev:04

5.0 Procedure of different condition monitoring techniques:

5.1 Vibration Monitoring and Analysis:

For Vibration measurements and fault diagnosis, following procedure is followed:

i. Overall Vibration Amplitude in terms of velocity and displacement in 3 principal directions i.e. Horizontal, Vertical & Axial are measured on periodic/ requirement basis.

ii. The schedules for equipment to be covered under the periodic vibration monitoring are prepared and uploaded in SAP system.

iii. Vibration data is maintained in the standard excel template in intranet and is uploaded in SAP system on regular basis.

iv. Equipment wise Vibration tolerance limits are prepared based on ISO standards, Equipment Baseline history and OEM guidelines.

v. Wherever vibration levels approach or exceed the tolerance limits, detailed vibration analysis is carried out to identify the root cause of vibration and the results of analysis along with action plan is informed to respective maintenance group.

vi. After the corrective action, Vibration measured and compared with vibration tolerance limits. Accordingly, clearance is given by MTP for continuous operation of the equipment

vii. For Turbine Generator, The LMI LMI/OD/OPS/MECH/007 'Turbine Generator Vibration Monitoring' is followed.

viii. Online vibration monitoring system is installed on all 4 Turbine generators and other critical rotating machines like PA, FD & ID Fans. The online system provides the operator alarm and trip indications.

ix. Responsibility of maintaining records of vibration data, trends and analysis reports lies with MTP department.

Responsibility: HOD (MTP)

TITLE	Maintenance Optimization Through Predictive Maintenance
LMI NO.	LMI/OIN/OPS/SYST/010 Rev:04

5.2 Infrared Thermography

Infrared Thermography is being used for condition monitoring of the following areas. The list of equipment to be covered in different areas under Infrared Thermography is prepared and maintained in SAP. Accordingly, their maintenance plans are uploaded and scheduled in SAP. IRT Data is maintained in local computer and also uploaded in SAP system.

AREA	FREQUENCY
1) 400KV switch yard, elements / joints and 21KV Bus ducts	Monthly
2) IRT of Power Transformers	Monthly
3) All HT Motors and Critical LT Motors of Main Plant and CHP	Monthly

Responsibility- HOD (EMD)

4) IR thermography for insulation survey	Yearly / Pre & Post

Responsibility- HOD (EEMG)

5.3 Dissolved Gas Analysis for Transformers:

Dissolved Gas Analysis of Transformers is being done as per the LMI/OGN/OPS/ELEC/019.

Responsibility – HOD (EMD)

5.4 Lube oil Monitoring & Wear debris analysis:

Lube oil analysis including wear debris is carried out and Wear Particle Concentration (WPC) is measured on periodic basis for the following equipment by O&M-Chemistry/NETRA. The analysis schedules are prepared by O&M-Chemistry. Equipment maintenance plans are uploaded and scheduled in SAP system accordingly.

1) Stage-I & II Coal mills Gearbox
2) Critical Coal Conveyors and Crusher Gearboxes.
3) Stage-I, & II PA, FD & ID Fan bearing lube oil.
4) Stage-I, & II Air Preheaters support and guide bearing lube oil.
5) MOT oil of all BFP’s and Main TG.
6) FRF of Main TG

Wherever PQ index is beyond limit, it is forwarded to NTPC R&D NETRA or to external agency for further detailed analysis.

Responsibility-HOD (Chemistry)

TITLE	Maintenance Optimization Through Predictive Maintenance
LMI NO.	LMI/OIN/OPS/SYST/010 Rev:04

5.5 Major Equipment Performance Monitoring:

Performance monitoring of Turbine Generator & Steam Generator is carried out by EEMG Department as per the schedule and records are maintained in EEMG Department.

Sl No	Parameter	Frequency of Monitoring
Steam Turbo-Generator		
1	Pressure Survey – Periodically measuring pressure along steam path (HP & IP)	Once in 6 months
2	Stand alone Trend monitoring of 1st stage pressure.	Monthly
3	HPT, IPT cylinder efficiency testing	Once in 6 months
4	Cooling Towers Efficiency testing	Once in a year
Steam Generator		
5	Dirty air flow tests for coal flow	Once in 6 months
6	APH Efficiency Test	Once in 3 months
7	Boiler Efficiency Test	Monthly
8	Maximum Tube Metal Temperature & The Corresponding Tube.	Daily Basis

Responsibility- HOD (EEMG)

5.6 Acoustic Steam leak detection system (ASLD):

Boiler tube leak is the major cause of unit outage in the area of ‘Boiler and Auxiliaries’. To detect tube leaks, operators have to monitor make up rate, furnace pressure fluctuations and listen for obvious noise during boiler operation. In contrast acoustic leak detection system can spot small leaks in boiler tubes, not detected by normal means.

As steam escapes through a hole in a boiler tube, it generates a broad band sound, the amplitude of which increases with leak size. ASLD systems are designed to detect the sound (2-8 kHz) of expanding steam as it escapes through even the smallest hole. The system consists of number of noise detection tubes (wave guides) mounted on the boiler at openings in the wall spread all over, sensitive microphones attached to the cold end of these tubes, preamplifiers that screen out background noise and a signal processing unit and data displays in the control room. An audio-visual alarm appears in the control room when a signal exceeds thethreshold set for the specific channel by utility personnel.

It is an advanced warning system to detect of non-catastrophic leak hours or days before they become large. The system has following advantages.

a) Damage to adjacent tubes is reduced, i.e. it helps in making timely repairs before leaks become serious.
b) Improved plant availability.
c) Maintenance and replacement cost is reduced.
d) It is possible to shut down the plant at a convenient time.
f) Increased personnel safety.

In Simhadri, Stage 1 Units are installed with ASLD system of M/s Sartech International, Chennai make and stage 2 Units are installed with ASLD system of M/s BHEL make. For early detection of tube failure in boiler monitoring of ASLD system is being done

Responsibility- HOD (Operation).

5.6.1 Valve Leak Detection

Valve seat leakage creates turbulence, generating noise levels and this is measured on sonic (100 Hz - 20 kHz) and ultrasonic (20 - 250 kHz) range of frequency and interpreted to detect seat leakage. Characterization of acoustic behavior of valve can further help in quantification of leakage flow. If we have a list of leaking valves detected with acoustics, these can be easily attended during small shutdowns such as during boiler tube leaks and can result in a lot of savings.

Acoustic Imager:

Acoustic imager works in ultrasonic frequency band and by integrating the sound source distribution data with the video image, the changing sound source is dynamically presented on the display screen. Acoustic Imager helps to quickly detect possible air, gas and vacuum leakage faults in noisy industrial environments.
The equipment is simple and convenient to operate and can be used quickly. It only needs to adjust two parameters, the test frequency range and test dynamic range to meet the vast majority of test requirements. Supports camera mode, video mode, and the data recording on the job site is flexible. Large areas can be scanned quickly which helps locate leaks much faster than with other methods. It also allows for filtering on intensity and frequency range

Advantages:

1. Detection and location of pressurized gas leakage and vacuum leakage
2. Partial discharge detection and localization
3. Structural vibration monitoring and localization

Responsibility- HOD (MTP)

5.7. Hydraulic oil Contamination monitoring:

Particle contamination reduces the service life of hydraulic components. Some level of particle contamination is always present in hydraulic fluid, even in new fluid. Contaminants circulating in fluid power systems cause surface degradation through general mechanical wear (abrasion, erosion and surface fatigue). This wear causes increase in number of particles and thus increasing the wear further. If this chain reaction of wear is not properly contained (by reducing the contamination) gaps grow larger, leakage oil flow increases in size and operating efficiency (e.g of Pumps, Cylinders) decreases. The Applicable standards for Oil Contamination Monitoring are NAS-1638- SAE AS 4059C and ISO 4406-1999. The minimum recommended cleanliness Level to be maintained is mentioned here under.

In hydraulic systems, 70% to 90% of wear and failures are contamination related only. Only 10 to 30% can be tracked back to misuse, defects or aging related. The experience shows that the contamination cannot be stopped, however, it can be slowed down. System efficiency can drop by up to 20% before an operator even detects a problem, such as cylinder drift, jerky/erratic operation or slower performance. The level of contamination or conversely, the level of cleanliness considered acceptable, depends on type of hydraulic system.

Type of Hydraulic System	Minimum Recommended Cleanliness Level	
	NAS 1638	ISO 4406
Silt sensitive	4	15/13/10
Servo	5	16/14/11
High Pressure (250 to 400 Bar)	6	17/15/12
High Pressure (150 to 250 Bar)	7	18/16/13
High Pressure (50 to 150 Bar)	9	20/18/15
Low Pressure (< 50 Bar)	10	21/19/16
Large Clearance	12	23/21/18

TITLE	**Maintenance Optimization Through Predictive Maintenance**
LMI NO.	**LMI/OIN/OPS/SYST/010 Rev:04**

In CHP, Hydraulic systems are existing in Stacker Reclaimers, Paddle Feeders, Wagon Tipplers and Side Arm Chargers. The hydraulic oil contamination monitoring in these equipment's is being done on regular basis and details are mentioned hereunder.

Equipment	**Type of Hydraulic system**	**Cleanliness level as per Standard-NAS 1638**	**Actual Cleanliness level Measured at an interval of 6 months**	**Remarks**
Stacker Reclaimers	Power pack With Servo System 68 Oil	NAS-7		
Wagon Tipplers	Power pack With Servo System 68 Oil	NAS-7		On-line filtration machines installed.
Side Arm Charger	Power pack With Servo System 68 Oil	NAS-7		On-line filtration machines installed.
Paddle feeders	Power pack With Servo System 68 Oil	NAS-7		

Responsibility- HOD (FM) / HOD (CHEMISTRY)

6.0 REVIEW:

HOD (MTP) will review & take the approval of competent authority (HOP). The document will be reviewed on 2 yearly basis or earlier, if required

7.0Annexure:

- Tolerance of different Condition monitoring measurements.
- Lube oil & Wear Particle Concentration Limits
- IRT tolerance Limits
- Transformer Oil & DGA Tolerance limits

TITLE	Maintenance Optimization Through Predictive Maintenance
LMI NO.	LMI/OIN/OPS/SYST/010 Rev:04

Annexure

Tolerance of different Condition monitoring measurements (Vibrations)

EQUIPMENT	Alarm Value Velocity in mm/s –RMS and Displacement in Microns Pk-Pk	Trip Value Velocity in mm/s –RMS and Displacement in Microns Pk-Pk
AHP IACs	11.00	18.00
AIR SIDE PUMPS SOS	4.50	11.00
ARCW PUMPS	7.50	11.00
ASH SLY PUMPS	9.00	18.00
BAHP PUMPS	9.00	18.00
CEP	9.00	14.00
CONDENSOR VAC PUMPS	7.50	11.00
CRUSHERS-CHP	11.00	18.00
ST-1 CW PUMPs	70 microns	100 microns 15 sec delay
ST-2 CW PUMPS	150 microns	225 microns 60 sec delay
ST-1 DMCW & ST-II ECW PUMPS	6.00	11.00
AHP FAHP PUMPs	7.50	18.00
FD FANs	7.50	18.00
FIRE HYD PUMPs & FIRE WATER BOOSTER PUMPS	9.00	18.00
H2 SIDE PUMP SOS	4.50	7.50
ID FANS	120 microns /4.5	200 microns /6.0
PA FANS	7.50	18.00
PAPH	9.00	18.00
PRIMARY WATER PUMPS	6.50	12.00
SAPH	9.00	18.00
SEA WATER PUMPS	9.00	14.00
TDBFP	7.50	18.00
MAIN TG SHAFT	200 microns pk-pk	300 microns pk-pk
MP-IAC	7.50	18.00
MP-PAC	7.50	18.00
DM PLANT POTABLE WATER PUMPs	7.50	18.00
SCN.A.FAN (AC & DC)	4.50	9.00
AHP BALP PUMPs	6.00	9.00
BOILER CC PUMPs	9.00	14.00

CONTROL FLUID PUMPs	6.00	11.00
AHP FLUSH WTR PUMPs	4.50	9.00
GSC BLR	6.00	11.00
SEAL AIR FANS	4.50	9.00
SWEET WATER PUMPs	7.50	18.00
AHP TACs	7.50	18.00
BFP- LOPs	4.50	9.00
CLINKER.GRINDERS	7.50	11.00
LP SEAL WTR PUMPS	4.50	9.00
MRHS COMPRSR	7.50	18.00
COAL MILLS	7.50	18.00
AHP- ESP VAC PUMPs	4.50	9.00
APH WASH PUMPs	4.50	9.00
DM MAKE UP PUMPs	4.50	9.00
FILTER WATER PUMPs	7.50	18.00
CC PUMPs	4.50	9.00
STACKER RECLAIMER BOOM CONVEYORS	11.00	18.00
SR- BUCKET-WHEEL	11.00	18.00
EFFNT.DIS.PUMPs	7.50	18.00
H2 PLANT COMPRESSOR	4.50	9.00
RAW WATER PUMPs	4.50	9.00
MDBFP	7.50	18.00
RAW COAL FEEDERS	15.00	18.00
ASH WTR R/C. PUMPs	7.50	18.00
COAL CONVEYORS	11.00	18.00
DEGSR PUMPs	4.50	9.00
HFO PR. PUMPS	4.50	9.00
LDO PR. PUMPS	4.50	9.00
HVAC PUMPS	4.50	9.00
GEN-TRANSFORMERS	11.00	18.00
T/S DRINK WTR PUMPS	7.50	18.00
MAIN TURB.EOP/DC	4.50	9.00
MAIN TURB.JOP/DC	4.50	9.00
MAIN TURB.JOP	4.50	9.00
TDBFP AOP	4.50	9.00
MAIN TURB.AOP	4.50	9.00

TITLE	Maintenance Optimization Through Predictive Maintenance
LMI NO.	LMI/OIN/OPS/SYST/010 Rev:04

Lube oil & Wear Particle Concentration Limits:

Sample	Frequency	Oil Type	Viscosity (CsT)	Moisture (PPM)	MI (PPM)	Acidity mg of KOH/gm of oil	PQ/ NAS
Turbine oil	Once in Week	Servo Prime 46	41.1-50.6	100	100	0.2	--
FRF(NAS value)	Once in Month	Fyrquel-N	41.1-50.6	1500	100	0.2	6
BFP's lub oil	Once in 15 day	Servo Prime 46	41.1-50.6	100	100	0.2	--
WT/SR lub oil	Once in 3 months				100		
APH support and guide bearing	Once in 03 Months	SERVOCYL-C680	--	--	1000	--	25
PA, FD & ID Fan bearing lube oils	Once in 03 Months	ISO-VG-68	61.8-74.8	100	100	0.2	25
Stage-I & II Coal mill Gearbox	Once in 03 Months	SERVO MESH SP 320	280-350	1000	1000	0.5	500

IRT tolerance Limits:

1	Infrared Thermography	Temperature of Joints/ Clamps/ Motor Terminals/ Equipment, etc.	Acceptable Limit: Up to 40 Deg C above ambient temperature.

Transformer Oil & DGA Tolerance limits:

	Moisture	Acidity	CO_2	C_2H_4	C_2H_2	C_2H_6	H_2	CH_4	CO
Limit as per IEEE C57.104	<10 PPM	0.2 mg KOH/ Gm oil	<2500 PPM	<50 PPM	<1 PPM	<65 PPM	<100 PPM	<120 PPM	< 350 PPM